



Simulation and experiment for dynamics of laser welding keyhole and molten pool at different penetration status

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Abstract

The penetration status profoundly influences the dynamic behaviors of keyhole and molten pool during laser welding. This paper presents a three-dimensional numerical simulation model with a ray-tracing method, which is dedicated to investigating the different penetration statuses in laser welding of type 304 stainless steel. In the proposed model, the factors include melting, evaporation, solidification, the vapor-induced plume heat transfer, buoyancy force, Marangoni effect, surface tension, and recoil pressure. Unlike existing similar work, three types of penetration status are investigated, including non-penetration, insufficient penetration, and full penetration. The weld cross-section geometry of simulation results is in good agreement with the experimental results. The behaviors of keyhole and molten pool, energy absorbed by the weldment, and flow field in the molten pool are discussed in details. It is found that the penetration status affects the dynamics of keyhole and molten pool, the temperature field, the absorbed energy, and the flow field, which is greatly connected with weld bead formation and joint quality.

Keywords Laser welding · Numerical simulation · Penetration status · Molten pool · Keyhole

1 Introduction

Nowadays, laser welding is widely applied in manufacturing due to its easy installation, small deformation, and high efficiency. Laser welding involves different penetration statuses such as non-penetration, insufficient penetration, full penetration, and excessive penetration [1]. The non-penetration laser welding leads to the lack of welding penetration, which results in cracks and porosities in the weld and severely reduces its strength [2]. In the case of insufficient penetration, the defects such as underfill [3] and root humping are easily generated. However, if the penetration status is excessive, the weld cap underfill would appear, which also reduces the strength of the welded component [1]. A lot of studies show that the weld defects are strictly related to the transient variations of the keyhole and molten pool. Different variations of the keyhole and molten pool, which are essential for the stability of the

welding process and welding quality [4, 5], correspond to different welding penetration. Therefore, it is necessary to study the relationship between penetration status and the dynamics of the molten pool and keyhole, which helps to optimize the parameters of the welding process and analyze the production of welding defects.

The transient variations of molten pool and keyhole are hot issues in the research because they are difficult to observe. Some researchers used an x-ray transmission on-line observation system to capture the variations of the molten pool and keyhole [6, 7]. However, the x-ray transmission on-line observation system is limited in wide application due to its high cost and harmful radiation. Several studies adopted a ‘sandwich’ method to observe the inner dynamic keyhole behavior, and the detailed introduction of the subject can be found in [8–10]. It is worthy to notice that the welding condition of ‘sandwich’ method is quite different from that in actual welding.

In recent years, the numerical simulation technology with the advantages of cheap and convenient is widely employed to understand the complex physical phenomena during laser welding. Some researchers have developed volumetric heat sources to investigate the molten pool and keyhole features [11], weld bead appearance [12], porosity formation [13], and spatter formation [14] during laser welding. However, the

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numerical methods using a volumetric heat source ignore the effect of Fresnel absorption which plays a key role in laser welding. In order to describe the laser welding process more precisely, researchers have proposed a heat source coupled with ray-tracing algorithms considering the effect of Fresnel absorption. Cho et al. [15] applied the VOF and a ray-tracing method to analyze the dynamic behaviors of the keyhole and molten pool. Lin et al. [16] investigated keyhole dynamics and the formation of keyhole-induced porosity considering the multiple reflection of laser rays in the formed keyhole. They found that the porosity formation experienced three steps, and interrupting one of them can suppress the pore formation. Zhang et al. [17] established a multi-physics numerical model to study the spatter formation. Pang et al. [18] proposed a 3D numerical model to study the influence of ambient pressure on the dynamics of deep penetration laser welding. However, the studies mentioned above mainly pay attention to the non-penetration or full penetration laser welding, and seldom research has been done to analyze the distribution characteristics of process parameters and the dynamics of the molten pool and keyhole under different penetration statuses. Thus, this study is devoted to investigating the connection between penetration status and welding process parameters, and the dynamic behaviors of the molten pool and keyhole in different penetration statuses.

In this work, three types of penetration status are investigated, including case 1 non-penetration (Fig. 1a), case 2 insufficient penetration (Fig. 1b), and case 3 full penetration (Fig. 1c). Laser welding experiments are implemented to analyze the distribution features of process parameters under three penetration statuses. A 3D numerical model is developed to analyze the absorbed energy and the dynamic behaviors of the keyhole and molten pool under different penetration conditions. Recoil pressure, Marangoni force, buoyancy force, and the vapor-induced plume heat transfer are taken into account in the simulation. Moreover, the dynamics of the molten pool and keyhole, the absorbed energy, and flow pattern under different penetration cases are compared and analyzed.

2 Experimental setup

In this study, the 5-mm thickness type 304 stainless steel plate is welded in experiments and simulations, and the chemical composition of the workpiece is listed in Table 1. The experiments used a fiber laser welding equipment (IPG YLS-4000) with a maximum power of 4 kW. The spot radius is 234 μm at the focal position, and the laser beam wavelength is 1070 nm. The welding head is installed on a KUKA robot. The sketch of the laser welding system is displayed in Fig. 2. In order to obtain the non-, insufficient, and full penetration welds, different welding parameters (laser power, defocus amount, and welding speed) were used, and some of them are listed in Table 2. The flow rate of argon shielding gas is the same, 20 L/min. Before welding, the acetone was used to remove the oil on the surface of specimens. After the experiment, all the prepared specimens were firstly ground using abrasive paper, then polished by polishing powder, and finally etched with aqua regia. The cross-sectional dimensions of the weld were measured based on the micrographs obtained by an optical microscope.

3 Mathematical modeling

3.1 Governing equations and assumptions

In the simulation model, the liquid in the weld pool is treated as an incompressible laminar flow with Newtonian viscosity. The governing equations, including mass conservation, momentum conservation (Navier-Stokes equation), energy conservation in Cartesian coordinates are expressed as follows [15, 19].

Mass conservation equation:

$$\nabla \cdot \vec{v} + M_s = 0 \quad (1)$$

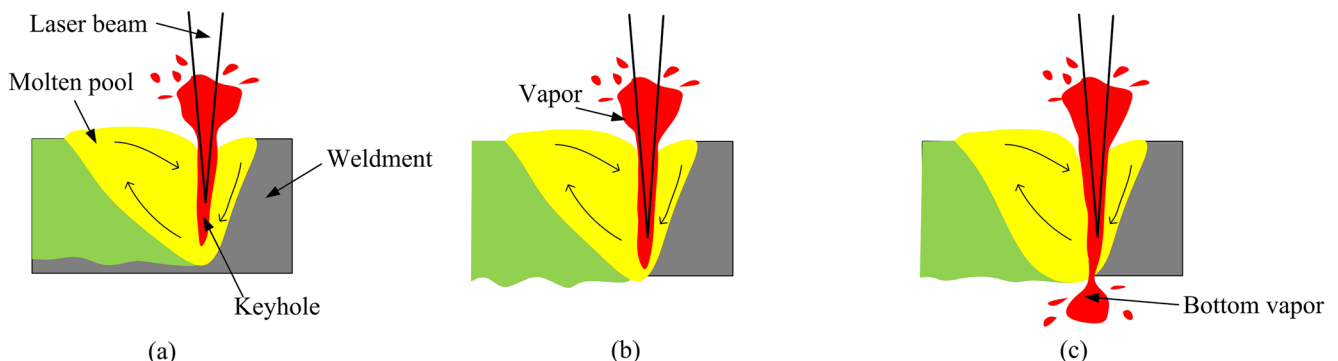


Fig. 1 The schematic of penetration statuses: **a** non-penetration, **b** insufficient penetration, **c** full penetration

Table 1 The material composition of 304 stainless steel

Element	C	Si	Mn	P	S	Ni	Cr	N	Fe
Content (wt. %)	0.047	0.46	1.06	0.034	0.005	8.24	18.16	0.048	Bal.

Navier-Stokes equation:

$$\frac{\partial \vec{v}}{\partial t} + \vec{v} \nabla \vec{v} = -\frac{1}{\rho} \nabla P + \nabla \cdot (\mu \nabla \vec{v}) - K \vec{v} + F_s \quad (2)$$

Energy conservation equation:

$$\frac{\partial h}{\partial t} + \vec{v} \nabla h = \frac{1}{\rho} \nabla \cdot (k \nabla T) + E_s \quad (3)$$

where $\vec{v}$, t , ρ , P , μ , K , h , k , and T are the vector of velocity, time, density, pressure, viscosity, drag coefficient, enthalpy, thermal conductivity, and temperature, respectively. The M_s is the source term of mass conservation equation, the F_s represents the source term of Navier-Stokes equation, and the E_s is the source term in energy conservation equation.

For tractability, the following assumptions were applied.

- (1) The laser energy conforms to Gaussian-like distribution, and the recoil force on the keyhole surface is calculated using the Clausius-Clapeyron equation.
- (2) Since the density of plume or gas is much lighter than solid or liquid stainless steel, it is assumed that the plume or gas area is empty. The bubbles in the weld pool are considered to be adiabatic bubbles, where the pressure is only related to volume. The metallic vapor generated during fiber laser welding is treated as a perfect gas with a temperature of 6000 K [15].
- (3) If the keyhole is blind, the velocity of vapor varies linearly from zero at the bottom of keyhole to the estimated velocity at the keyhole exit, as shown in Fig. 3a. When the keyhole is a through hole, it is assumed that the velocity of vapor increases linearly from 0 m/s at the

middle of the weldment to the estimated velocity at both keyhole exits, as shown in Fig. 3b.

3.2 Heat source model

In this work, a heat source is combined with ray-tracing to simulate the effect of Fresnel absorption owing to the multiple reflections of laser beam inside the keyhole. Therefore, it is assumed that the laser beam is composed of a number of energy rays. The original cell on the focal plane is divided into a series of sub-cells, each of which contains two sub energy rays having the same energy, as shown in Fig. 4. The total energy of the sub energy rays is equal to the energy allocated to the sub-cell. The allocated energy $q(x, y)$ of each sub-cell, which is determined only by the distance from the sub-cell to the focus center, is calculated by the Gaussian-like energy distribution function expressed as follow.

$$q(x, y) = \frac{2Q}{\pi R^2} \exp(-2(x^2 + y^2)/R^2) \quad (4)$$

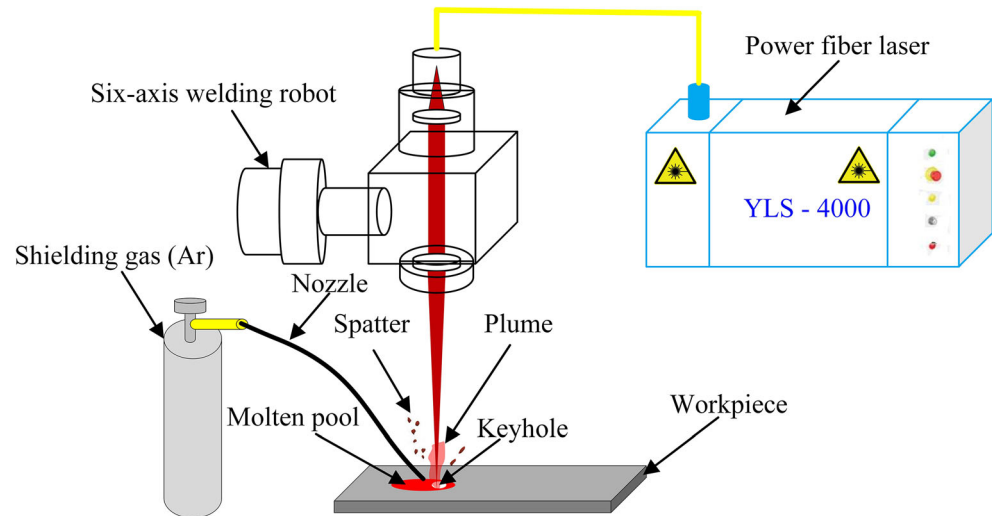
where Q is the input laser power, R is the focus radius. The incident direction of each energy ray is based on the diffraction parameters [20].

The laser absorptivity on the specimen depends on Beer-Lambert law and the absorption energy coefficient of the material. The details of heat transfer on the free surface, absorption of the laser beam, method of calculation vapor velocity, and buoyancy force are given in Cho's research [15]. The algorithm of free surface tracking combined with Level-Set and VOF methods is used. A comprehensive presentation of

Table 2 Some of welding parameters used in the experiments

Penetration status	Laser power (W)	Welding speed (mm/s)	Focus position (mm)
Case 1	3600	40	+ 1
	3400	40	0
	3600	45	- 1
Case 2	3600	35	+ 1
	3700	41	0
	3600	40	- 1
Case 3	3600	20	+ 1
	3500	15	0
	3600	25	- 1

Fig. 2 The schematic of the laser welding system



laser beam ray-tracing method and free surface tracking is given in previous researches of Na's team [21, 22].

3.3 Boundary conditions

The boundary conditions are complex and important due to various complicated physical phenomena in laser welding. The detailed description of boundary conditions is as below.

The side surfaces of the computational domain are defined as the continuative condition, and the top and bottom surfaces are set as the outflow condition.

For upper and bottom surfaces of weldment, heat convection and heat radiation are considered. The thermal boundary is expressed in Eq. (5).

$$k \frac{\partial T}{\partial \vec{n}} = -h_c(T - T_0) - \varepsilon_r \sigma (T^4 - T_0^4) \quad (5)$$

where $\vec{n}$ is the normal vector of the surface, h_c is the convection coefficient, T_0 is the environmental temperature, ε_r is the radiation coefficient, and σ is the Stefan-Boltzmann constant. With regard to the reheated effect of metallic vapor, the ambient temperature is set to 6000 K in region 1 and region 2, as shown in Fig. 5. The coefficients r_1 , r_2 , and r_3 are the radii of

region on the free surface that is affected by the metallic vapor. The temperature in the other regions is the room temperature of 298 K.

The energy on the keyhole wall is balanced between the absorbed energy, thermal radiation and convection, and the energy loss caused by material evaporation [15, 20]. The energy balance equation is calculated as

$$k \frac{\partial T}{\partial \vec{n}} = q_a - h_c(T - T_1) - \varepsilon_r \sigma (T^4 - T_1^4) - q_e \quad (6)$$

$$q_e = m_{\text{evap}}(c_p T + H_v) \quad (7)$$

$$m_{\text{evap}} = \lambda P_r \sqrt{\frac{1}{2\pi R_0 T}} \quad (8)$$

$$P_r = A p_0 \exp\left(H_v \frac{T - T_v}{R_0 T T_v}\right) \quad (9)$$

where q_a is the absorbed energy, T_1 is the metallic vapor temperature, and q_e is the loss of energy due to metal evaporation. The m_{evap} here is the evaporation rate which can be obtained from the modified Langmuir equation as shown Eq. (8). c_p is the liquid-specific heat, and H_v is the latent heat of vaporization. The adjustment factor λ can be set to 0.82, P_r is the recoil

Fig. 3 The variation of the velocity of vapor: **a** The blocked keyhole, **b** the through keyhole

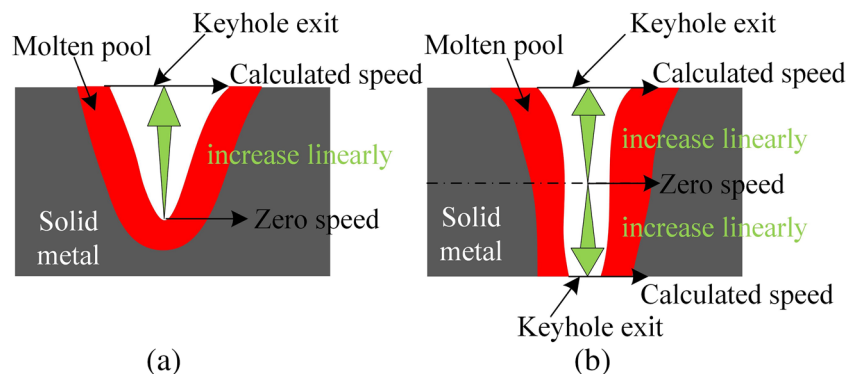
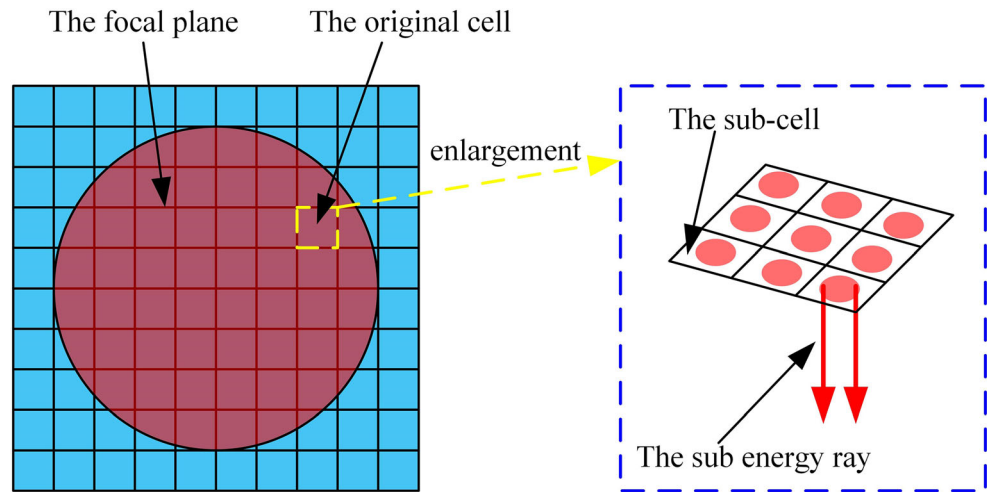


Fig. 4 Discretization of the laser beam

pressure, and R_0 is the universal gas constant. Further, A is a coefficient which can be set to 0.54, P_0 is the atmospheric pressure, and T_v is the boiling temperature.

The pressure boundary condition on the keyhole wall is described as

$$P = P_r + P_\gamma + P_s + P_d \quad (10)$$

where P_r is the recoil pressure that can be calculated using Eq. (9), P_γ is the surface tension, P_s is the hydrostatic pressure, and P_d is the hydrodynamic pressure.

The surface tension P_γ is calculated by [15, 23]

$$P_\gamma = \gamma/R_c \quad (11)$$

$$\gamma(T) = 1.87 - 4.3 \times 10^{-4}(T - 1727) - 1.3 \times 10^{-8} \times RT \times \ln \left[1 + 0.00318 f_s \exp \left(\frac{1.66 \times 10^8}{RT} \right) \right] \quad (12)$$

where γ denotes the surface tension coefficient which evaluates as a function of temperature $\gamma(T)$, and R_c is the radius of keyhole surface curvature. R is the gas constant, and f_s represents the sulfur concentration.

The hydrostatic pressure P_s and hydrodynamic pressure P_d are calculated as

$$P_s = gh_1 \quad (13)$$

$$P_d = \frac{2u^2}{g} \quad (14)$$

where g represents the gravity, h_1 denotes the depth of the weld pool, and u is the velocity of the liquid.

3.4 Numerical implementations

Three different simulations were conducted to investigate different penetration statuses in laser welding. Figure 6 displays the schematic of a type 304 stainless steel plate and a computational domain. The blue centered zone is equally divided into delicate cells of 0.1 mm in size, and other zones are gradually divided into large cells. The thermophysical material properties are shown in Table 3. As shown in Table 4, the welding parameters are given for each simulation. To reduce the computational cost, the simulation time is set to 300 ms.

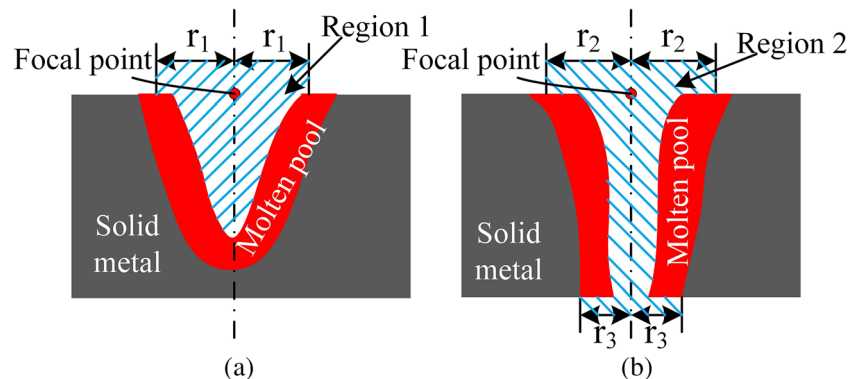
Fig. 5 The schematic of region affected by the metallic vapor: **a** the blocked keyhole, **b** the through keyhole

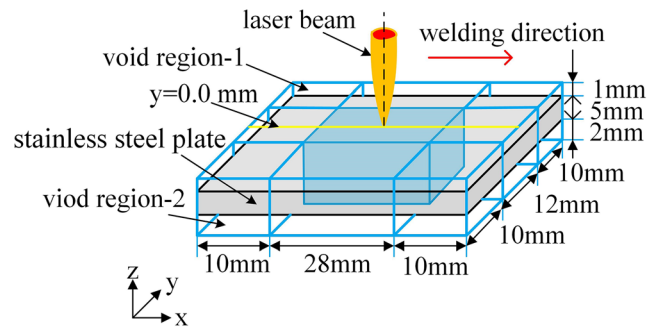
Table 3 The thermal properties of the 304 stainless steel

Properties	Values
Solid density	7350 kg/m ³
Liquid density	6900 kg/m ³
Viscosity	0.006 kg/m s
Surface tension	1.87 N/m
Surface tension gradient	-0.43×10^{-3} N/m K
Specific heat of solid	720 J/kg K
Specific heat of the liquid	800 J/kg K
Latent heat of fusion	2.61×10^5 J/kg
Latent heat of vaporization	6.36×10^6 J/kg
Solids temperature	1633 K
Liquids temperature	1727 K
Boiling temperature	3000 K
Thermal conductivity of liquid phase	24 W/(m.K)

4 Results and discussions

4.1 Experimental results

The experimental results are shown in Fig. 7, including case 1 non-penetration, case 2 insufficient penetration, and case 3 full penetration. It is obvious that these three cases present very different weld appearances on the top and bottom surfaces. At case 1 non-penetration, there exhibits small spatters on the top surface, but no weld formed at the bottom surface owing to non-penetration. And the shape of cross section is like the nail weld shape. At case 2 insufficient penetration, it was found underfill on the top surface and humping on the bottom surface, which are unacceptable defects. And the shape of cross section is also similar with the nail weld shape. At case 3, it can be observed that the width of the weld bead on the top and bottom surfaces are larger than that of the other two cases, meaning the full penetration was achieved. However, the shape of cross section is not the nail weld shape; it is more like a hyperbolic shape. In order to reveal the relationship between the welding process parameters and penetration statuses, all experimental samples were analyzed. The process parameters of each type of penetration status are transformed into line energy and plotted in Fig. 8, where the power

**Fig. 6** The schematic of the stainless steel plate and computational domain

density D_p is calculated by Eq. (15), and the line energy L_e is calculated by Eq. (16). It is confirmed that if the line energy is less than 90 J/mm, the non-penetration will be achieved. When the line energy is over 145 J/mm, the full penetration occurs. However, the case 2 insufficient penetration occurs when the line energy is between 90 and 145 J/mm.

$$D_p = Q/S \quad (15)$$

$$L_e = Q/v_w \quad (16)$$

where Q is the laser average power, S is the contact area of laser beam on the workpiece, and v_w is the welding speed.

To further study the distinctions between the three cases, it is necessary to analyze the transient behaviors of the weld pool and keyhole. Thus, simulations of every case were conducted.

4.2 Validation

To verify the rationality of the simulation model and algorithm, the cross-sections of simulated welds were compared with the experimental results. To exhibit the welded zone, the weldments were cut, polished, and then etched. Figure 9 exhibits the welded zone of the experimental and simulated results in the three penetration statuses.

The contour of the experimental welded zone is put on the simulated weld. The consequence indicates that the simulated shapes are consistent with experimental results. Also, the width of top weld bead (TW), the width of middle weld bead (MW), the width of bottom weld bead (BW), and the depth of weld bead (DH) were measured in both simulated results (SR) and experimental results (ER). The measured results are

Table 4 The welding parameters of the simulation

Welding parameters	Non-penetration	Insufficient penetration	Full penetration
Laser power	3.6 kW	3.7 kW	3.7 kW
Beam radius	0.234 mm	0.234 mm	0.234 mm
Focus position	0 mm	0 mm	0 mm
Welding speed	45 mm/s	38 mm/s	20 mm/s
Line energy	80 J/mm	97.37 J/mm	185 J/mm

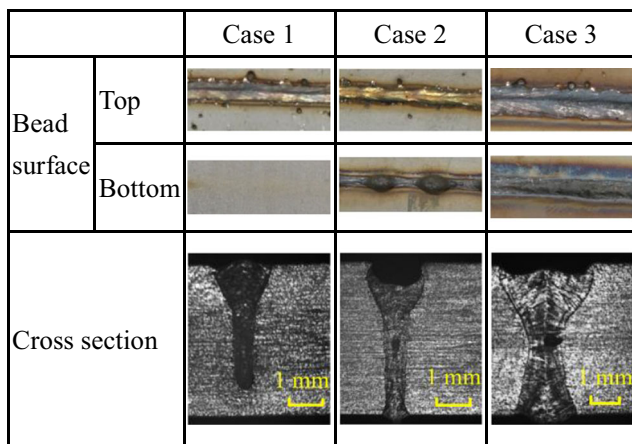


Fig. 7 The experimental results of three cases of penetration

tabulated in Table 5, showing that the simulated results match experimental values accordingly. Thus, the developed simulation model in this research is rational.

4.3 Discussion

4.3.1 The dynamics of the molten pool and keyhole

Figure 10 shows the transient variations of the keyhole and molten pool in different penetration statuses during the laser welding. At case 1, as shown in Fig. 10a, it is observed that the size of the molten pool increases from $t = 10$ ms to $t = 145$ ms, which is induced by the increase in energy absorbed in the initial stage. After that, the energy absorbed by the weldment keeps relatively steady, so the weld pool stays almost unaltered over time. It also can be found that the workpiece has not been penetrated, and the upper part of the melt pool is longer than the lower part. The reason for this is that when the

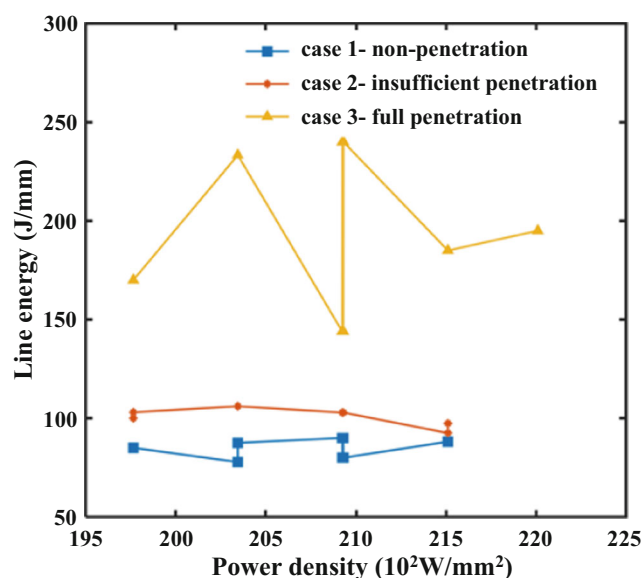


Fig. 8 Process parameter distribution of three cases of penetration

workpiece is unpenetrated, there is a huge temperature gradient on the top surface of the melt pool, resulting in a big Marangoni force. The result is the fluid on the top surface of the melt pool flows along the reverse welding direction, which makes the upper part of the melt pool become longer. Also there is a clockwise vortex at the bottom of the melt pool, which can be found in Fig. 13a. Due to this clockwise vortex, the heat at the bottom of the molten pool will dissipate faster. Meanwhile, the solidification at the bottom of the melt pool is relatively faster because of being surrounded by a solid region with low temperature and high thermal conductivity. Thus, the upper part of the melt pool is longer than the lower.

The behaviors of the keyhole and molten pool in case 2 are displayed in Fig. 10b. It is observed that this case is completely different from case 1 regardless of the shape of the weld pool or the behavior of keyhole. Initially, the weld pool is similar to that of case 1. However, when the plate is penetrated, the appearance of the molten pool looks kind of like a '>'. Simultaneously, the weld pool at the bottom is longer than the upper part, and the middle part is the smallest. As for a reason, it can be explained as follows: when the workpiece is penetrated, a temperature gradient exists on the top and bottom surfaces of the weld pool, so the Marangoni effect produces, leading the molten metal flows to the rear of the weld pool. It is clearly seen that the temperature difference on the bottom surface is greater than that on the top surface, and therefore, the molten metal on the bottom surface (greater Marangoni force) flows more strongly to the rear of the weld pool than that on the top surface. Besides, the molten metal on the top and bottom surfaces of the weld pool are exposed to the air, which leads to delay in solidification, and the middle part is cooled quickly because of being surrounded by a solid region with low temperature and high thermal conductivity. Therefore, the weld pool at the bottom is longer than the upper part.

Figure 10c exhibits the transient behaviors of the keyhole and weld pool of case 3. When the plate is penetrated, the characteristic of the molten pool is a little similar to that of case 2. However, the length of the upper part of the molten pool is equal to that of the bottom. The main reason is that when the process is full penetrating, the more metal vapor produces, and then the vapor drag force on the top rear keyhole wall increases enhancing the backward flow. So, the strength of backward flow on the top surface caused by the Marangoni force and the vapor drag force is almost equal to that on the bottom surface induced by the Marangoni force, and finally the length of the upper part of the molten pool is the same as that of the lower. In case 3, the melted metal on the front wall of the keyhole is more than in the other two cases due to the raise of line energy. Surface tension on the bottom surface of the weld pool makes the molten metal flow upwards. However, the hydrostatic pressure and hydrodynamic pressure exert downward pressure on the bottom surface of the

Transverse cross section

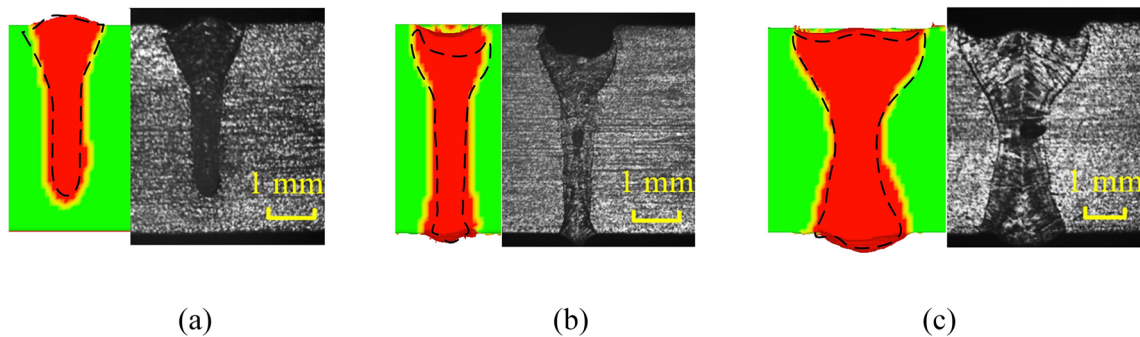


Fig. 9 Comparison of the simulation and experimental results: **a** case 1 non-penetration, **b** case 2 insufficient penetration, **c** case 3 full penetration

weld pool. When the sum of the hydrostatic pressure and hydrodynamic pressure exceeds the surface tension, the bulges on the bottom surface form.

The front keyhole wall maintains almost vertical, and the rear keyhole wall varies severely in laser welding process, as shown in Fig. 10a–c. The variation of rear keyhole wall is mainly caused by the recoil pressure which is unevenly distributed on the rear keyhole wall. To further study the behavior of keyhole, the evolutions of keyhole radius and depth in the three simulations are analyzed and depicted in Fig. 11. In the beginning, the keyhole depth and radius increase rapidly as the absorbed energy increases. Then, the variations of keyhole radius and depth keep relatively stable after some point, which indicates that the laser welding process is relatively stable from that moment. The keyhole radius curves reveal that the case 3 leads to keyhole with a large opening, which indicates that there is a large recoil pressure on the keyhole inlet. Because of the large recoil pressure, the top rear keyhole wall moves backwards, forming a large inclination angle. The large inclination angle is helpful to prevent spatter formation at the top rear keyhole wall, which has been found in [24] work. The keyhole depth curves show that the keyhole is blind in most time in case 2, while the keyhole often opens at the bottom in case 3. Also, the keyhole depth has higher fluctuation in case 2 and case 3 compared with case 1. The large undulation of keyhole depth indicates that the keyhole is easy to collapse. When the keyhole is through, the laser beam

escapes from the through keyhole, leading to the decrease in absorbed energy and recoil pressure. Hence, the pressure on the keyhole wall is unbalanced, and the keyhole tends to collapse, and finally the keyhole depth decreases. The keyhole profiles in Fig. 10b, c also exhibit a little higher fluctuation compared with the keyhole profile in Fig. 10a.

4.3.2 Evolution of absorbed energy by the weldment

Every energy ray was traced to calculate the energy absorbed by the weldment, and the results are displayed in Fig. 12. The absorbed energy curves show that the absorbed energy in case 3 has higher undulation compared with the other two cases. At case 3, the bottom of keyhole is often opened, which can be observed in Fig. 11c. When the through keyhole is produced, the absorbed energy decreases rapidly because a number of laser rays escape through the keyhole without interacting with the weldment. Besides, it can be found in Fig. 11b that the bottom of the keyhole is opened at some moments. Hence, the absorbed energy curve in case 2 shows a little higher undulation compared with that of case 1. For comparison purpose, the ratios of absorbed energy of the three cases are obtained and shown in Fig. 12d. The ratios of absorbed energy of the three cases are 89.75%, 86.86%, and 71.70%, respectively. It reveals that the ratio of absorbed energy is related to penetration status and is the smallest in case 3. Therefore, when the process is full penetrating, a portion of the total laser energy is wasted.

4.3.3 Flow field in the molten pool

To investigate the flow field of the weld pool, the streamline is plotted. The streamlines of the longitudinal velocity field in cases 1, 2, and 3 at $t = 240$ ms, 245 ms, and 250 ms are calculated and displayed in Fig. 13a–c, respectively. In case 1, the molten pool displays two main types of the vortex located in specific regions. There is a counterclockwise vortex inside the upper part of the molten pool, which is induced by

Table 5 The dimension of weld bead profile

Variable	Case 1		Case 2		Case 3	
	SR	ER	SR	ER	SR	ER
TW/mm	1.42	1.88	1.43	1.59	2.83	2.81
MW/mm	0.76	0.65	0.75	0.66	1.11	0.93
BW/mm	—	—	1.16	0.84	2.03	1.92
DH/mm	3.87	3.75	5.00	5.00	5.00	5.00

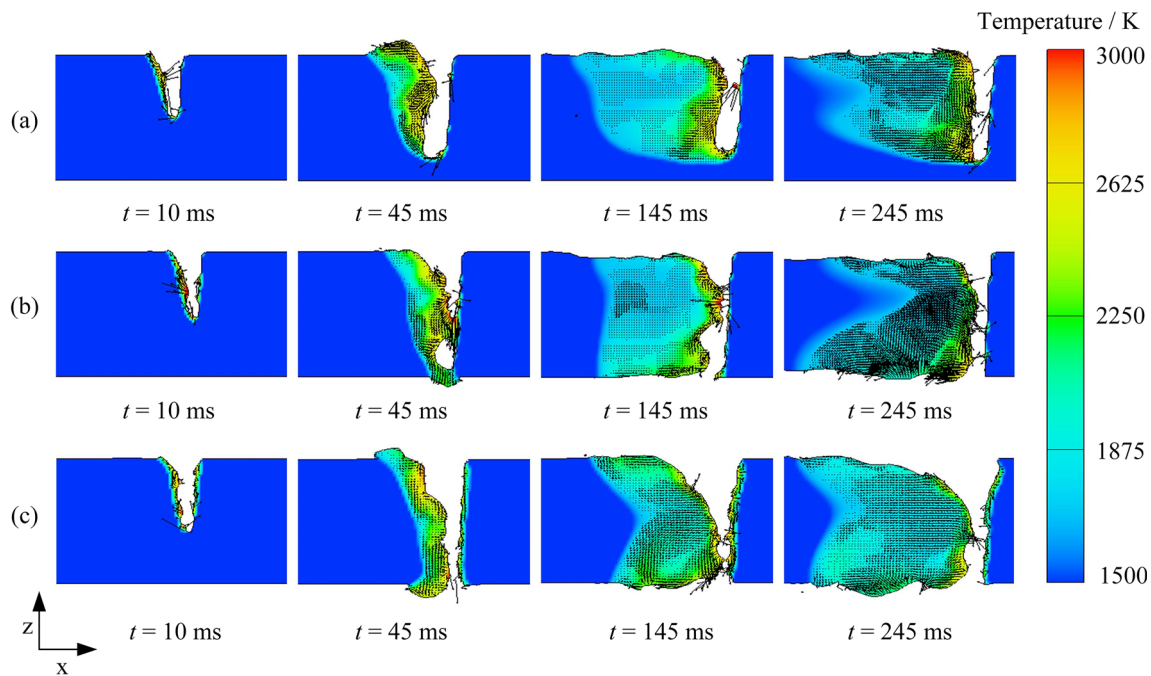


Fig. 10 Temperature field on the longitudinal cross section [$y = 0.0$ mm]: **a** case 1 non-penetration, **b** case 2 insufficient penetration, **c** case 3 full penetration

the effect of surface tension gradient, and the fluid exhibits a large and full vortex inside the lower part of the melt pool. It shows in Fig. 13a that the fluid at the bottom of the molten pool flows backward due to recoil pressure, and then flows upwards. When the upward fluid meets the downward fluid of the upper part, the fluid flows forwards. Also, the fluid at the

rear of the keyhole moves downwards which is caused by the surface tension, recoil pressure, and hydrostatic pressure. Hence, a large and full vortex at the lower part of the melt pool is formed. The streamline in case 1 stays relatively steady over time, which is associated with the small fluctuation of absorbed energy. The flow type at the upper part in case 2 is

Fig. 13 The streamline and velocity magnitude color map of the longitudinal cross section [$y = 0.0$ mm]: **a** case 1 non-penetration, **b** case 2 insufficient penetration, **c** case 3 full penetration

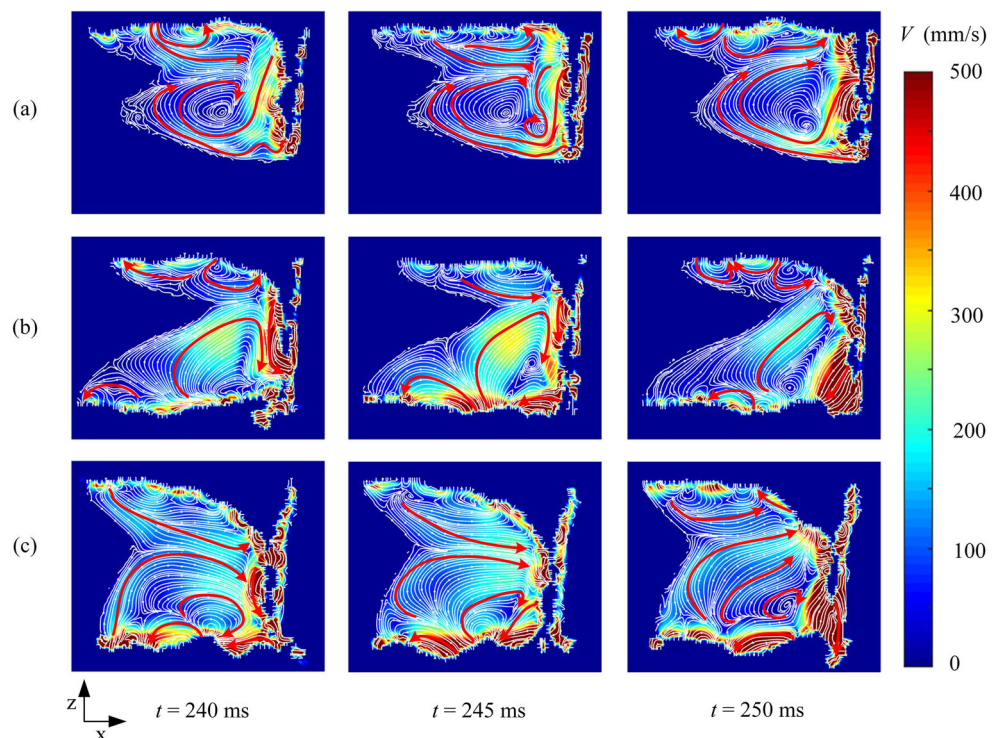
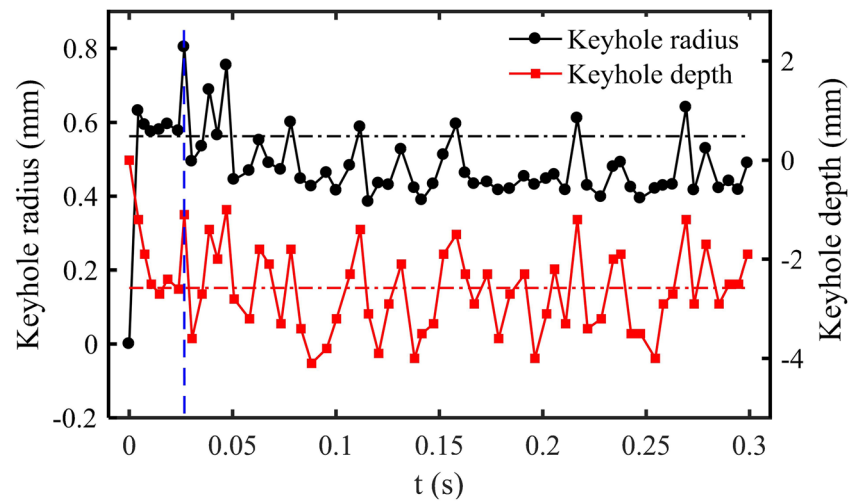
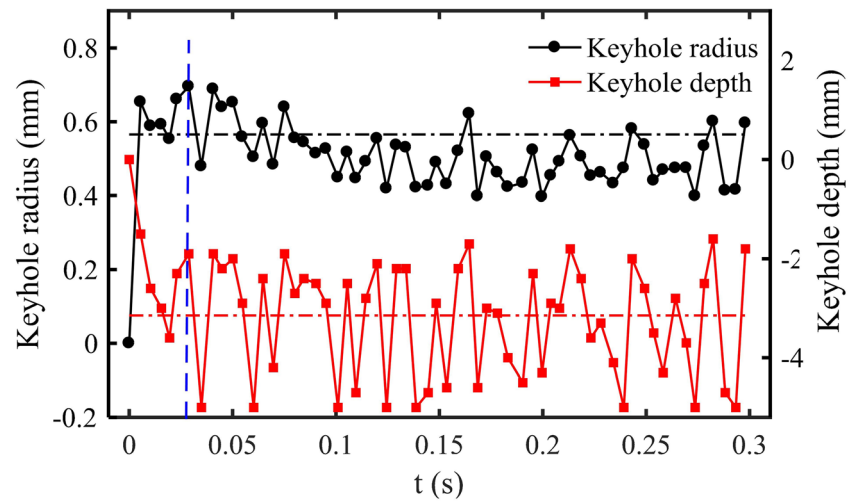


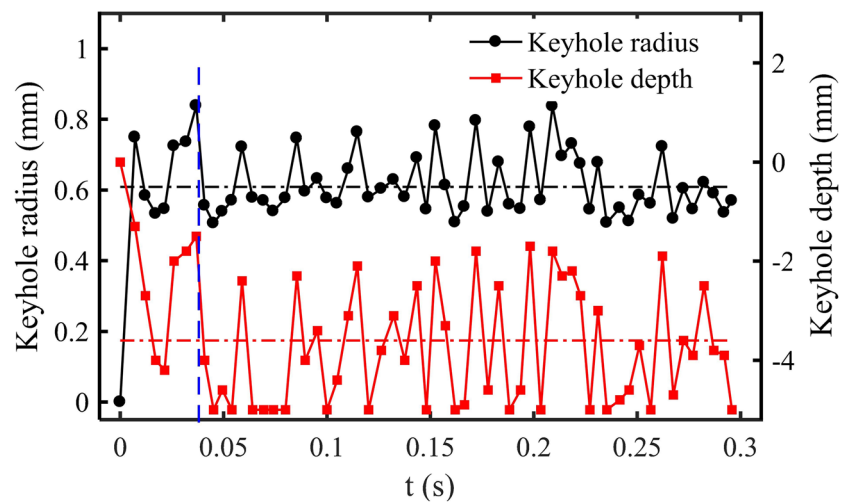
Fig. 11 Variation of keyhole radiuses and keyhole depths. **a** Case 1 non-penetration, **b** case 2 insufficient penetration, **c** case 3 full penetration



(a)



(b)



(c)

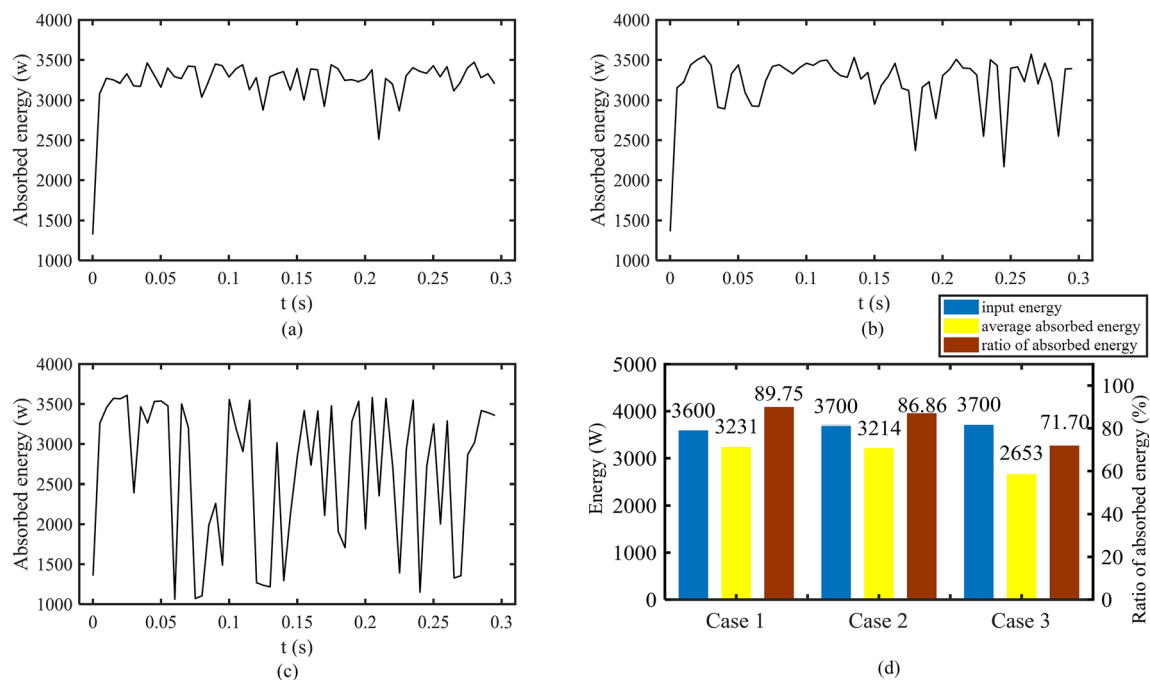


Fig. 12 Variation of absorbed energy and ratio of absorbed energy in the three simulations: **a** case 1 non-penetration, **b** case 2 insufficient penetration, **c** case 3 full penetration, **d** ratio of absorbed energy

almost as same as case 1. The distinguishing feature of flow pattern in case 2 is that some fluids at the bottom of the weld pool flow backward, and the others flow upwards to form a clockwise vortex. The backward flow is caused by a combination of surface tension, gravity, and Marangoni force. The upward flow is mainly associated with the temperature distribution of the molten pool. Results of case 3 shown in Fig. 13c are quite different from that of the other two cases. The melted metal at the rear of the weld pool bottom moves upwards along the solid-liquid interface until meeting the downward fluid. The main reason is that the temperature of the melted metal on bottom surface is higher than that of the fluid inside the weld pool, which can be observed in Fig. 10c, so the molten metal moves upwards. With the effect of Marangoni force and a big backward force caused by the vapor on the top surface of the weld pool, an anticlockwise vortex is formed inside the upper part of weld pool, as shown in Fig. 13c.

5 Conclusion

A 3D numerical simulation model has been developed to investigate the dynamics of molten pool and keyhole in different penetration status during laser welding of 304 stainless steel. The key findings can be summarized as follows:

- (1) The penetration statuses are highly related to the input of line energy. Full penetration can be achieved when the line energy is over 145 J/mm. If the line energy is less than 90 J/mm, non-penetration will occur. The

insufficient penetration can be obtained when the line energy is between 90 and 145 J/mm.

- (2) The dynamics of molten pool and keyhole in every penetration status has its characteristics. The Marangoni force, the rate of metal solidification, and the vapor drag force play a key role on the shape of a weld pool.
- (3) The energy absorbed by the weldment in full penetration has higher undulation in comparison of the other two penetration statuses because some laser rays escape from the through keyhole.
- (4) The flow fields under three types of penetration status are entirely different. The fluid field in non-penetration is relatively steady and regular, but the flow fields in insufficient and full penetration are more complicated. The factors influence the flow field of a weld pool, including the Marangoni force, the recoil pressure, the vapor drag force, and the temperature distribution inside the weld pool.

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Data availability Not applicable.

Compliance with ethical standards

Consent to participate Not applicable.

Consent to publish Not applicable.

Conflict of interest The authors declare that they have no conflict of interest.

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